Sparse Impulsive-Noise Removal: Classical Detection and Repair versus Neural Speech Enhancement

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Repository: https://github.com/zhi0708/elec5305-project--540686380-

Project site: https://zhi0708.github.io/elec5305-project--540686380-/

Abstract

Sparse impulsive interference presents a preservation problem: removing a short event can damage considerably more speech than the event itself. This project compares event-localised DSP with an official pretrained GTCRN enhancer using controlled insertions into a reference speech recording. An energy detector and a three-feature STFT detector drive local attenuation or linear interpolation. The experiment uses 90 development mixtures and 180 held-out mixtures, covering five event templates, three interference levels and three acoustic locations. Exact insertion intervals are available for every case. The STFT detector achieves event F1 0.362, compared with 0.080 for energy detection, while retaining low recall. GTCRN reduces mean event reconstruction error more strongly (5.87 dB), but introduces greater clean-region alteration than the STFT local methods. These results support a restoration-versus-preservation trade-off, not a universal ranking of DSP and neural enhancement.

1. Research question and motivation

The research question is: for sparse short-duration impulsive interference in speech, how does event-localised DSP detection and repair compare with a modern pretrained neural speech enhancer in event restoration, speech preservation and computational cost? A secondary question concerns interference overlapping an existing speech transient.

Predictive reconstruction [1], stationary-wavelet suppression [2] and HHT detection [3] provide historical context for specialised impulsive-noise processing. The current project implements neither the wavelet nor HHT algorithms. Recent sparse adaptive filtering [5] motivates examining disturbance sparsity and localisation: its author repository discusses sparse optimisation and the consequences of altering undisturbed samples. It is used as motivation rather than reproduced or claimed as a baseline. GTCRN [4] supplies an accessible lightweight learned comparator with official pretrained weights. The project contribution is the controlled experiment, transparent student-developed detector and separation of event error from collateral alteration.

2. Data and controlled corruption

The existing repository contains one 33.62-second, mono 8 kHz speech recording. It is OSR_us_000_0010_8k.wav from the Open Speech Repository, verified against a byte-identical public mirror. Its source and SHA-256 are recorded in data/speech_manifest.json. It is treated as a controlled reference rather than a certified noise-free corpus. Only the first 30 seconds are used. They form six non-overlapping five-second clips: clips 1-2 are development data and clips 3-6 are test data. The split separates time segments, not speakers. Upsampling to 16 kHz meets GTCRN's input requirements but does not restore missing bandwidth. All speech clips share one fixed scaling that limits the original recording peak to 0.15; mixtures and outputs are never independently normalised.

Two synthetic templates are seeded Gaussian bursts with Hann envelopes, lasting 5 and 20 ms. Three real templates are cropped from ESC-50 [6]: door knock (31.375 ms), mouse click (16.125 ms) and keyboard event (18.312 ms). A 2 ms RMS envelope locates the strongest peak; the crop expands while above 10% of peak, with a 75 ms limit on each side, 2 ms guards and short boundary fades. Source files, crop times, individual attribution and file hashes are provided. There is only one extracted template per real class, so this tests three examples rather than environmental-event diversity.

Each template is added separately at a pause, a steady voiced candidate and a speech-transient candidate. Locations are derived from the reference waveform: minimum smoothed energy identifies the pause; autocorrelation with an energy gate and low-flux preference identifies steady voiced speech; large positive spectral flux identifies the transient. These are acoustic proxies, not human-verified vowels or phoneme-aligned /p/, /t/ and /k/ releases. The reference is used for constructing controlled cases and evaluating outputs, never supplied to a tested detector.

Event levels are -6, 0 and +6 dB. Level means event RMS relative to the RMS of a local speech window extending 50 ms on each side of the inserted interval. In near silence, that reference has a floor at 10% of whole-clip RMS. An insertion mask covers the exact half-open interval [start, end), including its fades. The generator retains clean speech, added event, corrupted speech and the mask. All case intervals are saved in the manifest; scripts/export_case.py materialises any case. Float arrays are used for measurements to avoid clipping or quantisation. The grid contains 2 x 5 x 3 x 3 = 90 development mixtures and 4 x 5 x 3 x 3 = 180 test mixtures. These are controlled variants of a small reference, not 270 independent utterances.

3. Detection and local restoration

The analysis uses 8 ms windows and 2 ms hops, with a Hann window for spectra and a 256-point real FFT. Short frames favour transient timing; zero padding samples the spectrum more densely but does not increase physical frequency resolution. Short windows improve temporal localisation at the expense of frequency discrimination. Energy describes local intensity, spectral flux captures positive frame-to-frame spectral change, and bandwidth describes spectral spread. Suppression is evaluated separately from speech preservation; LPC reconstruction remains a possible extension.

The energy baseline compares a local energy rise against a threshold. Its rise is 10 log10((frame_energy + floor)/(local_median_energy + floor)). The median spans 101 frames, approximately 202 ms; the floor is 1% of the recording's 60th-percentile frame energy. This adaptive baseline can detect sudden increases but can mistake normal speech onsets for interference. Its runtime excludes all FFT calculations.

The STFT method combines three features: positive energy rise, positive spectral flux and spectral bandwidth. Flux is the sum of squared positive changes between consecutive L2-normalised

magnitude spectra. Its robust z-score uses the recording median and a scale equal to the maximum of 1.4826 times the median absolute deviation, 0.1 times its standard deviation, and a numerical floor. Bandwidth is the square root of the power-weighted frequency variance. The score is max(energy_rise,0)/6 + max(flux_z,0)/3 + bandwidth/1500. The coefficients are fixed heuristic scales, not learned from test labels. Their benefit is interpretability; no ablation is available to isolate each feature's contribution.

Candidate frames exceeding the threshold are mapped to sample intervals, including the frame's half-window support. Candidates longer than 150 ms are rejected. The same duration rule applies to both detectors. The development search considers seven energy thresholds and eight STFT thresholds. It maximises event F1 with false events on the two clean development clips counted 45 times, giving each clean clip the same weight as its 45 noisy variants. This makes preservation an explicit selection preference. The frozen thresholds are 9 dB energy rise and STFT score 5. The test set is not used to choose either threshold.

Attenuation applies a gain of 0.2 inside the predicted event, with up to 2 ms entry and exit ramps. Interpolation replaces the predicted samples with a straight line connecting the preceding and following samples. At file boundaries, the missing neighbour is zero. Both methods preserve the input exactly outside the predicted mask. This guarantee concerns the predicted interval; false detections can still alter genuinely clean speech. An oracle interpolation control uses the true mask solely to diagnose reconstruction limitations and is excluded from detector performance and runtime comparison.

4. Neural comparison and evaluation protocol

GTCRN uses the official DNS3 checkpoint, unmodified model architecture and a pinned source commit. No model is trained or fine-tuned. The adapter follows the official 512-sample FFT, 256-sample hop and square-root Hann analysis/synthesis convention. Modern PyTorch complex STFT tensors are converted into the model's real/imaginary input representation and converted back for iSTFT. Output length is explicitly matched to input length. There is no case-specific gain correction, delay fitting or reference-guided enhancement.

Event detection uses one-to-one matching of predicted and true intervals with positive overlap. TP, FP, FN, precision, recall and F1 are pooled over test cases. A partial overlap counts as an event hit, so sample-level metrics are also stored to reveal poor interval coverage. Onset error is the absolute start-time difference for matched events only; it does not summarise misses. Reports separate pause, steady voiced and transient locations, and retain event type and strength in per-case CSVs.

Event RMSE compares output with the clean reference inside the true insertion interval. Collateral RMSE uses every sample outside that true interval, thereby counting false-positive repair and any neural alteration. Event error reduction is 10 log10(noisy event MSE / output event MSE); positive values mean improvement. Whole-file SI-SDR and STOI [7] are secondary checks, not substitutes for local measurements. Table values are arithmetic means across cases; median RTF is reported separately. A mean of dB improvements is not equivalent to a dB ratio of mean errors. Clean-input preservation is evaluated over all four test clips (20 seconds) with no inserted events.

Both methods run on the same INTEL(R) XEON(R) PLATINUM 8573C CPU environment. Neural inference is CPU-only with one PyTorch thread. Model loading and one warm-up call are excluded. Local runtime includes detector computation and repair; neural runtime includes STFT, model and iSTFT. RTF is wall-clock processing time divided by five seconds of audio. It measures offline throughput, not causal latency: local median filtering and recording-wide feature normalisation use future context. No streaming implementation is claimed.

5. Measured results

5.1 Detection

Detector	Precision	Recall	F1	TP / FP / FN	Mean matched onset error (ms)
ENERGY	0.047	0.289	0.080	52 / 1062 / 128	3.77
STFT	0.521	0.278	0.362	50 / 46 / 130	2.76

The STFT detector markedly increases precision and overall F1 but does not improve recall. Both methods miss most of the weak events. The STFT method has 130 missed events out of 180; it is a conservative detector, not a reliable general impulse remover. Its sample-level recall is only 0.087, lower than event recall, because some detections overlap only a fraction of an inserted event. Inference from event F1 alone would therefore overstate useful localisation.

Acoustic location	Detector	Precision	Recall	F1
silence	energy	0.003	0.017	0.005
silence	stft	0.545	0.300	0.387
speech transient	energy	0.080	0.517	0.139
speech transient	stft	0.579	0.367	0.449
steady voiced	energy	0.052	0.333	0.091
steady voiced	stft	0.400	0.167	0.235

Figure 1 shows precision, recall and F1 against strength separately for these locations. Full grouped values also separate the five event templates. The strongest performance is template-dependent: the short synthetic click and real mouse click are detected more often than the cropped knock and keyboard event. This is consistent with the chosen spectral features favouring abrupt broad-spectrum change, but without an ablation the explanation remains a hypothesis. The transient condition is not uniquely hardest in this small experiment, so the experiment does not establish consonant-overlap conditions as the most difficult category.

5.2 Restoration, preservation and computation

Method	Event RMSE	Clean-region RMSE	Event error reduction (dB)	SI-SDR (dB)	STOI	Median RTF
Unprocessed	0.016776	0.000000	0.00	32.95	0.971	0.00000
Energy + interpolation	0.013971	0.004933	1.25	12.13	0.877	0.00030
STFT + attenuation	0.014943	0.000026	0.97	33.77	0.972	0.00183
STFT + interpolation	0.014994	0.000062	1.15	33.64	0.974	0.00183
GTCRN (DNS3 pretrained)	0.006374	0.002243	5.87	17.89	0.899	0.01649

GTCRN produces the lowest measured mean event RMSE and about 5.87 dB mean event error reduction, compared with 1.15 dB for STFT interpolation. However, STFT interpolation has far lower collateral RMSE (0.000062 versus 0.002243) and preserves global speech metrics more closely. GTCRN's mean whole-file STOI is lower than the unprocessed sparse-corruption input

despite better local event repair. This shows why event removal alone does not establish whole-speech benefit. It may also reflect domain mismatch: the reference originates at 8 kHz and is upsampled, whereas the model is a general 16 kHz enhancer. The experiment cannot isolate domain mismatch from learned processing effects.

The energy baseline has particularly high collateral error because its false events repair many legitimate speech intervals. Locality alone does not guarantee preservation if the detector is wrong. STFT attenuation and interpolation yield modest average improvement and a zero median improvement: at least half of cases receive no useful repair. The oracle control yields mean event RMSE 0.015971 and mean event error reduction 2.85 dB. Even perfect event location does not guarantee that a straight line reconstructs oscillatory speech. Its event RMSE can remain large on speech overlaps despite exactly zero collateral error.

Figure 2 displays event error against collateral error. Figure 3 uses the same 20 ms synthetic event waveform at the same absolute amplitude in all three locations; it is a separate demonstration from the local-RMS strength sweep. The fixed amplitude is set from the +6 dB steady-voiced example and is not rescaled in silence or at the transient. Waveform and spectrogram panels use identical inputs for all methods. Figure 5 shows the measured runtime trade-off. All measured median RTFs are below one, but this is an offline CPU result. GTCRN parameter tensors occupy 192,980 bytes; the checkpoint is 579,819 bytes. The model has 48,245 parameters including fixed ERB mappings, of which 23,669 are trainable. These are model-storage figures, not measured process peak memory; DSP stores no learned weights but does use waveform, frame and FFT work buffers.

5.3 Clean-input false alarms and preservation

Method	False events / 20 s	False events / minute	Mean clean RMSE	Modified clean samples
Energy + interpolation	24	72	0.00492323	3.5575%
STFT + interpolation	1	3	0.00000564	0.0403%
GTCRN (DNS3 pretrained)	N/A	N/A	0.00221101	100.0000%

The STFT detector produces one false event over the clean test clips, versus 24 for the energy baseline. Figure 4 shows that STFT false alarm in clip 6: a small waveform change coincides with a sharp normalised spectral-flux peak, rather than an unusually loud pulse. This explains how abrupt spectral evolution can cross the threshold even at low absolute energy. The waveform has not been phoneme-labelled, so the event is described as a legitimate reference-speech change, not identified as /t/ or /k/. The clean no-op test also shows that GTCRN changes almost all samples; the changes are numerically measurable, not evidence that every altered sample sounds worse. Audio examples enable informal inspection, but no formal listening study or perceptual preference claim is made.

6. Limitations and conclusions

The study has one source recording, four test clips and three real-event templates. Development and test share recording conditions, and mixing variants are dependent. No speaker-generalisation claim, confidence interval based on independent utterances or significance test is justified. The original recording is attributed to the Open Speech Repository and verified against a byte-identical public mirror; it is not independently certified noise-free. Location proxies do not establish phoneme identity. Event matching permits partial overlap; sample-level metrics expose that limitation. The chosen detector prioritises avoiding collateral alteration and misses weak or less broadband events. Straight-line repair is weak for longer voiced intervals. Neither a learned impulse-specific model nor the recent sparse adaptive algorithm is tested, and peak memory and causal latency are not measured.

Within these controls, STFT features reduce false detections compared with energy-only detection, while pretrained GTCRN offers stronger local restoration and greater whole-waveform alteration. The strongest supported conclusion is that sparse corruption requires separate evaluation of event reconstruction, clean-region preservation and computation. A future study should expand source speakers and event templates, label consonant releases, and investigate either LPC reconstruction or detector-gated neural replacement. Neither extension is needed to interpret the completed core experiment.

7. Reproduction and implementation acknowledgement

Run: python -m pip install -r requirements.txt; python scripts/prepare_data.py; python -m pytest -q; python scripts/run_experiment.py; python scripts/build_deliverables.py. The run creates manifest intervals, frozen thresholds, per-case detection/restoration results, no-op tests, figures and exact example arrays. scripts/export_case.py exports any controlled input; scripts/enhance.py processes a user-supplied WAV without reference labels. See README for platform-specific environment and website instructions.

AI assistance was used in code implementation and documentation. Reported measurements were produced by executing the accompanying experiment code, not taken from expected outcomes or substituted from published GTCRN benchmark numbers.

References

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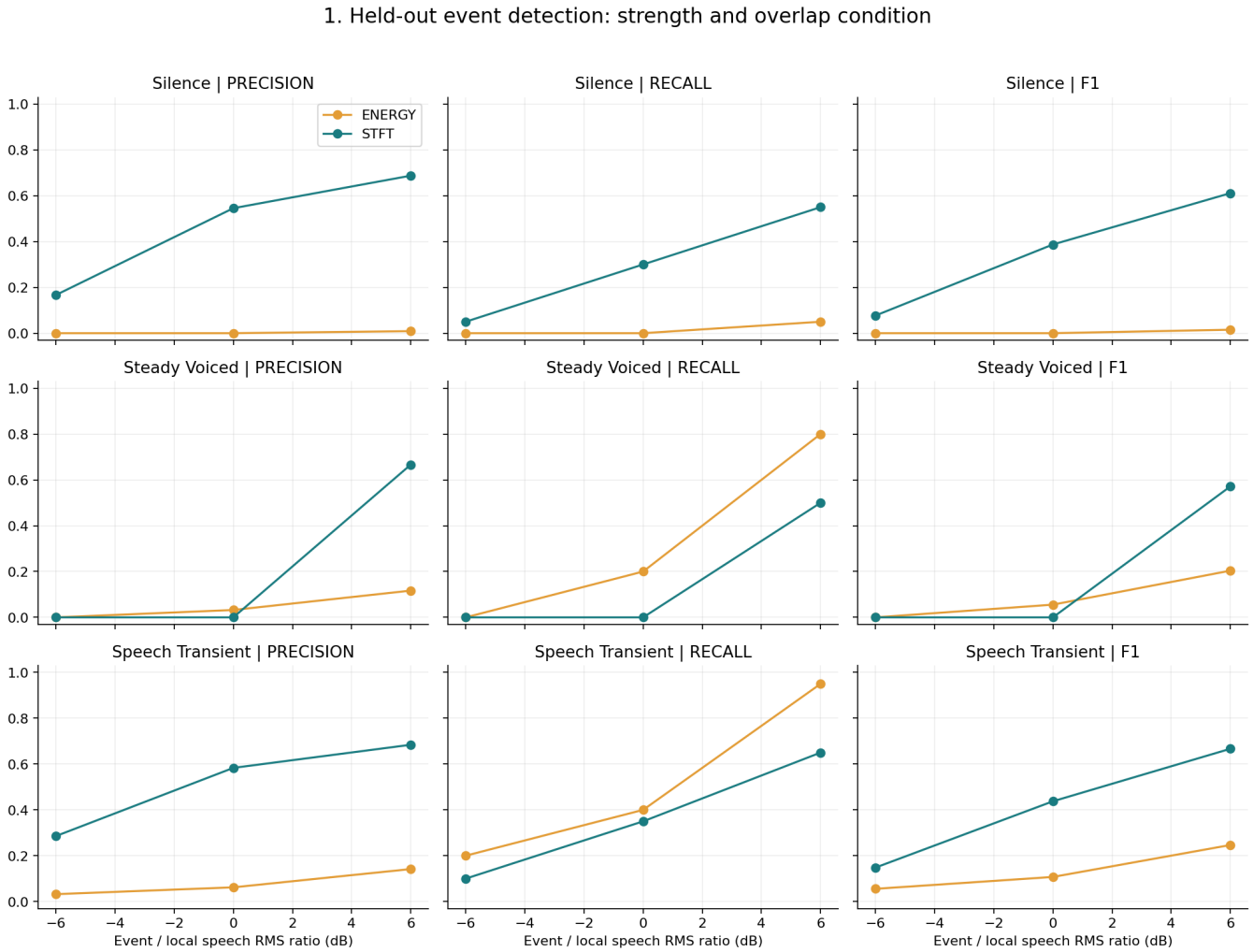
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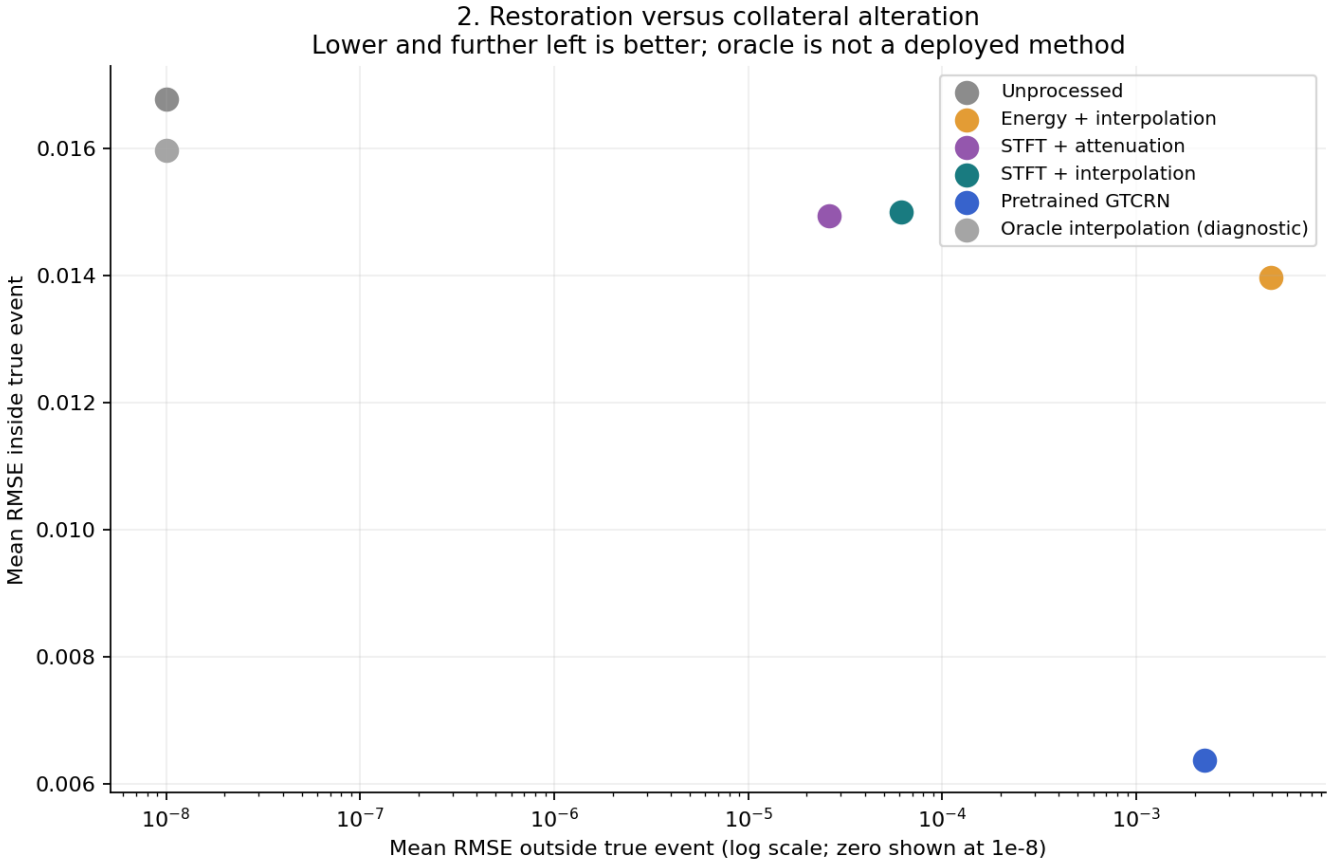
7. C. H. Taal et al., "An Algorithm for Intelligibility Prediction of Time-Frequency Weighted Noisy Speech," IEEE TASLP, 2011. STOI implementation: https://github.com/mpariente/pystoi

Figure 1. Detection by strength and acoustic location



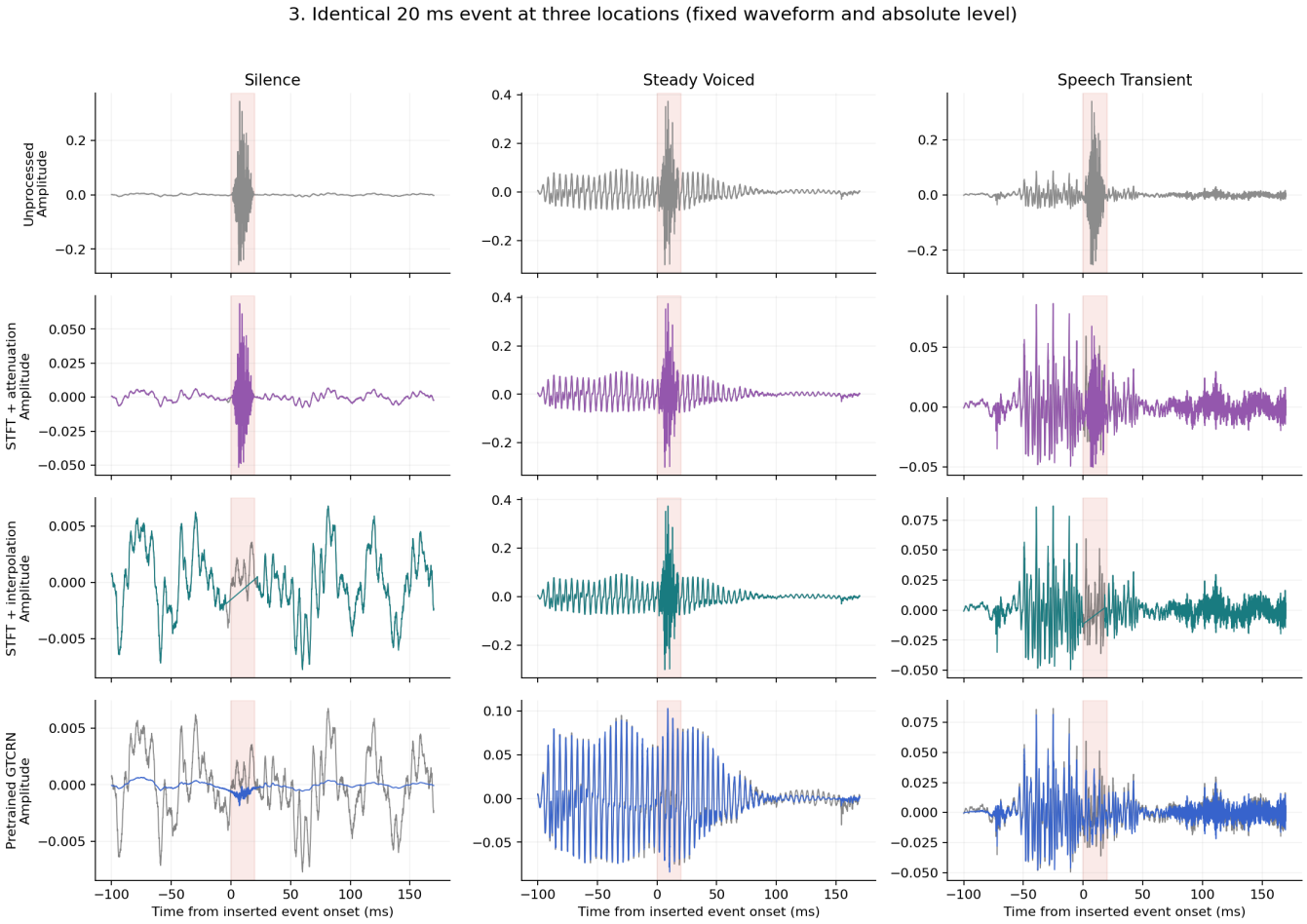
Held-out event metrics; all event templates pooled within each overlap/strength condition.

Figure 2. Event restoration versus collateral alteration



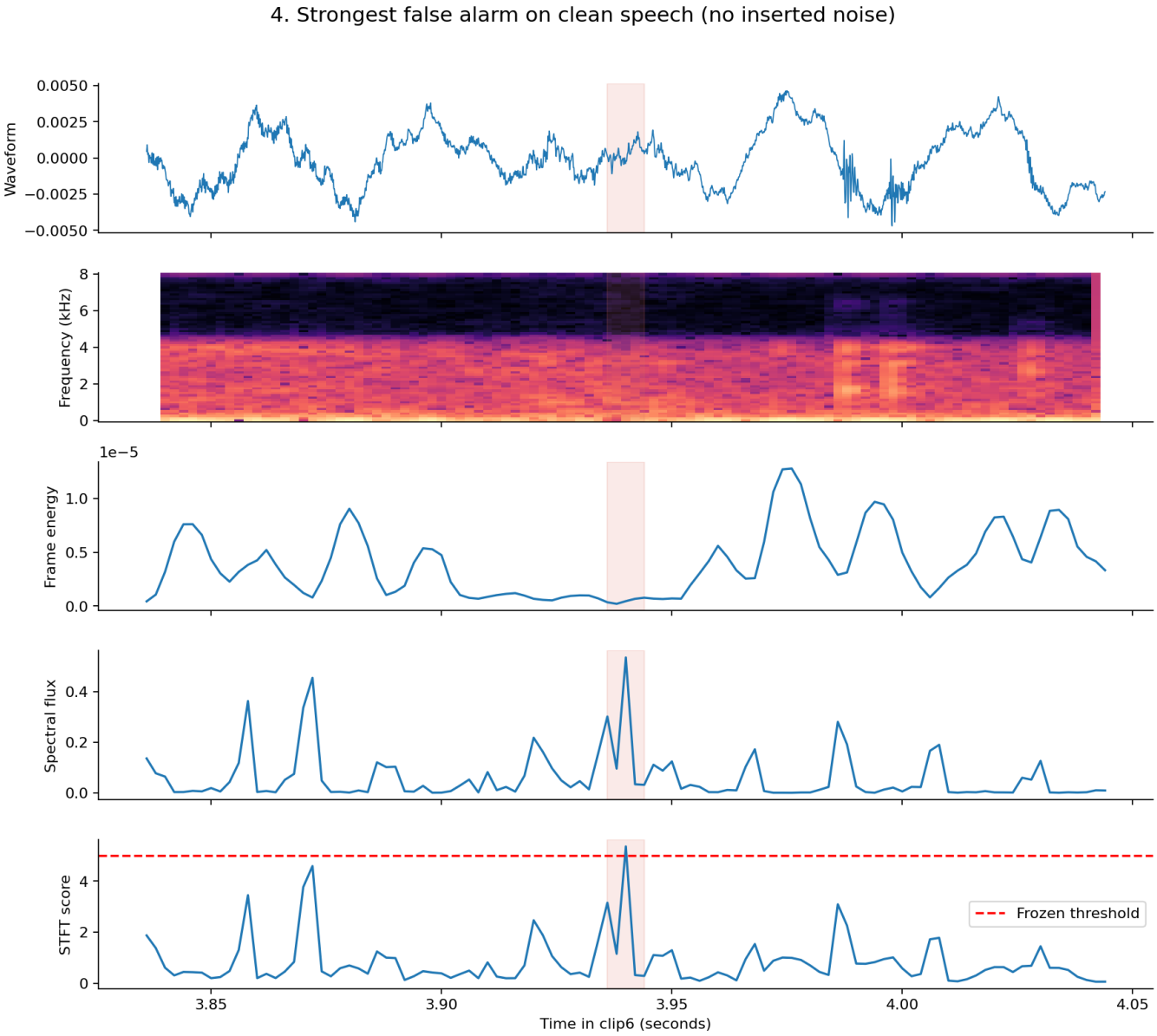
Lower error on both axes is desirable. True-mask interpolation is diagnostic only.

Figure 3. Identical event at different locations



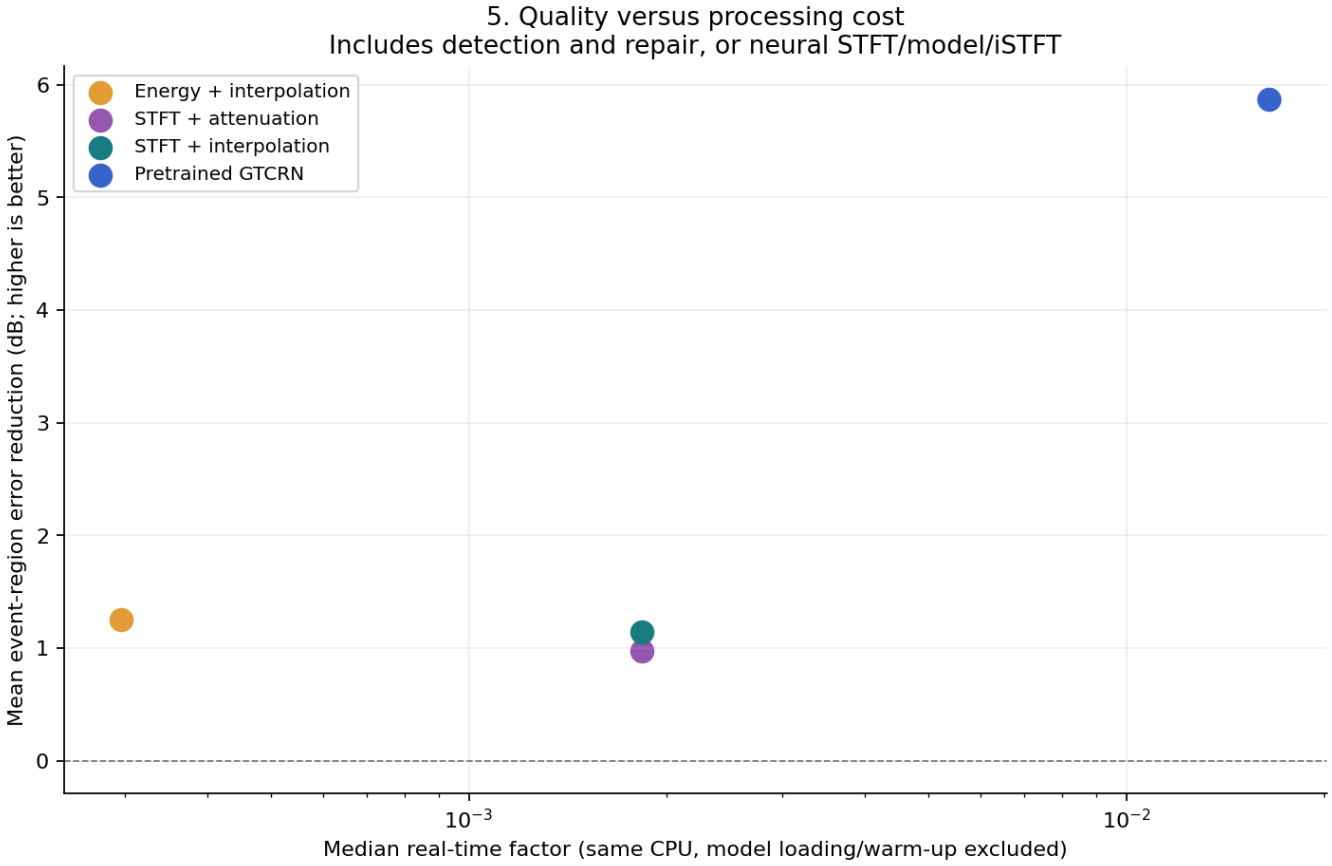
A fixed 20 ms event waveform and absolute amplitude are used for all three insertions.

Figure 4. False detection on clean reference speech



The high normalised flux peak triggers an event despite low absolute energy. No phoneme label is available.

Figure 5. Restoration quality and CPU runtime



Processing is timed on the same CPU. Model loading and warm-up are excluded; no causal latency is inferred.